

From this equation, we should see why the unity-gain frequency, f_{un} , is called the *gain-bandwidth product*. Note that for every increase in frequency by 10 (decade) above f_{3dB} , we get a decrease in A_{OL} by 10 (–20 dB). Alternatively, for every increase in frequency by 2 (octave) above f_{3dB} , we get a decrease in A_{OL} by 2 (–6 dB).

The closed-loop bandwidth (which we'll call f_{3dBCL}) of the op-amp circuit cannot be larger than the bandwidth of the op-amp. Therefore, to estimate the bandwidth of a closed-loop op-amp circuit, assuming the op-amp is the limiting factor, we can write

$$A_{CL} \cdot f_{3dBCL} = f_{un} = \text{gain-bandwidth product} \quad (24.34)$$

If the op-amp circuit has a closed-loop gain of one (a follower configuration as seen in Fig. (24.7)), the bandwidth is f_{un} . The op-amp in Fig. 24.21 used in a follower configuration would have a bandwidth of (roughly) 100 MHz. If this op-amp were used in a closed-loop configuration with a gain of 10, then the bandwidth of the op-amp circuit would be estimated as 10 MHz.

24.2 An Op-Amp with Output Buffer

In the previous section, the low-frequency open loop gain of the op-amps in Figs. 24.8 or 24.21 was determined by a product of the diff-amp's gain, A_1 , and the common-source's gain, A_2 or

$$A_{OLDC} = A_1 \cdot A_2 = \overbrace{g_{mn}(r_{o2} || r_{o4})}^{A_1} \cdot \overbrace{g_{mp}r_{o7}}^{A_2} \quad (24.35)$$

which, from the simulations (see Fig. 24.19 for example) was 1,000. The small-signal resistance on the output of the op-amp, r_{o7} , is (from Table 9.2) 333 k Ω . If we were to connect a 10 k Ω resistor to the op-amp output, then A_{OLDC} would drop to 33. As seen in Fig. 24.1, we can add a buffer to the output of second stage to isolate it from a load resistance (or a large capacitance). We could use a source-follower (that has a voltage gain of 1), as seen in Figs. 21.46 or 21.48. However, this addition limits the output swing of the op-amp (which may be OK in some situations). Next we could try using a push-pull topology, as seen in Fig. 21.49. However, the voltage gain of the push-pull amplifier is >1 (see Eq. (21.116)) so if we were to include it in our basic op-amp of Fig. 24.21, we would have a three-stage op-amp (meaning all three stages have voltage gains greater than one). A three-stage op-amp can be challenging to compensate over production corners (and temperature). Also, as seen in Eq. (21.116), a small load resistor (k Ω) would still kill the push-pull amplifier's gain. To keep the gain high, let's try to use a first-stage topology with a large gain (increase A_1) while using a push-pull amplifier for the second stage. The gain of the second stage can still drop but, since the gain of the first stage is large, we still end up with a reasonable overall gain.

Towards the goal of increasing the first-stage gain, consider the op-amp schematic seen in Fig. 24.29. For the first stage, we've used the cascode diff-amp originally seen in Fig. 22.30. The gain of this diff-amp (now A_1) is 500 (from Tables 9.2, and 20.1 and Eq. (22.48)). The second stage of the op-amp is the topology seen in Fig. 21.50. Its gain depends on the load resistance it drives. If no load is connected to its output, then

$$A_2 = -10 \cdot (g_{mn} + g_{mp}) \cdot \frac{(r_{op} || r_{on})}{10} = -31.6 \text{ V/V} \quad (24.36)$$

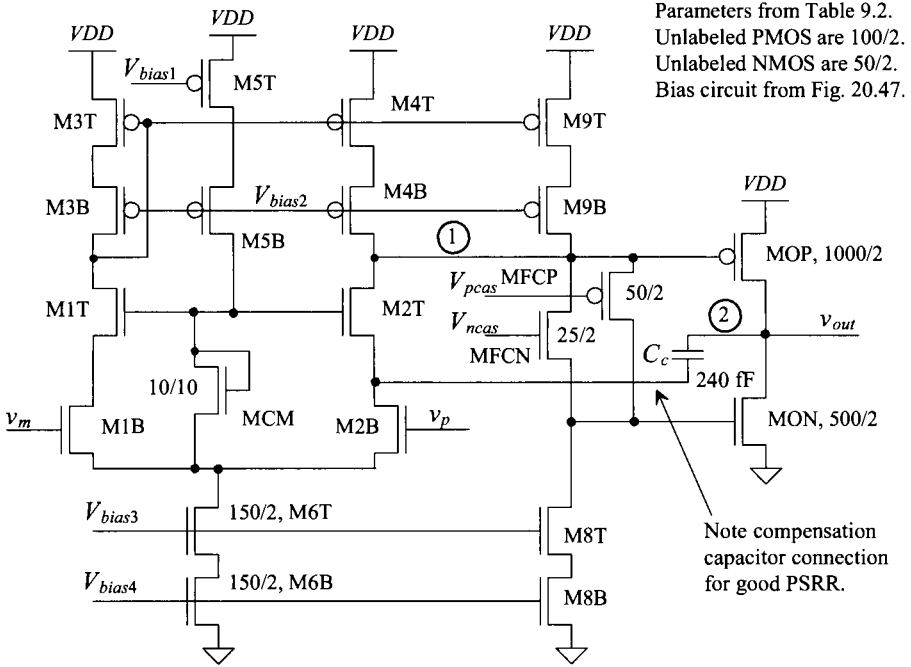


Figure 24.29 A CMOS op-amp with output buffer.

giving an overall low-frequency gain, A_{OLDC} , of 15,800 ($= 84$ dB). Note that the widths of MON and MOP are ten times wider than the values specified in Table 9.2 and so, as seen in Fig. 21.50, they conduct 100 μ A of current. In Eq. (24.36) we multiplied the g_m s in Table 9.2 by 10 ($g_{mon} = 10 \cdot \beta_n (V_{GS} - V_{THN}) = 10 \cdot g_{m, Table 9.2}$) and divided the output resistances given in Table 9.2 by 10 ($r_{omon} = \frac{1}{\lambda \cdot 10 \cdot 10 \mu A} = \frac{r_{on}}{10}$).

Compensating the Op-Amp

We increased the gain of the first stage by increasing R_1 (the resistance at node 1 is now set by cascoded current sources). As seen in Eq. (21.65), this pushes the pole lower in frequency. As just described in Eq. (24.33), this decrease in f_1 ($= f_{3dB}$) and increase in low-frequency gain *doesn't have any effect* on the gain-bandwidth product ($= f_{un}$) of the op-amp. What this indicates, to a certain extent, is that we can still compensate the op-amp with a 240 fF capacitor as we did in the last section. To feed the indirect compensation current to node 1 we connect the compensation capacitor to the source of M2T, as seen in Fig. 24.29. This ensures good PSRR.

The open-loop response of the op-amp in Fig. 24.29 (without a load) is seen in Fig. 24.30. The phase-margin is roughly 70 degrees. The step response of the op-amp driving a 1k resistor and a 10 pF capacitor is seen in Fig. 24.31. The input signal is a pulse from 100 mV to 900 mV. Since the gain of the amplifier is -1 and the common-mode voltage is 500 mV, the output swings from 900 mV to 100 mV.

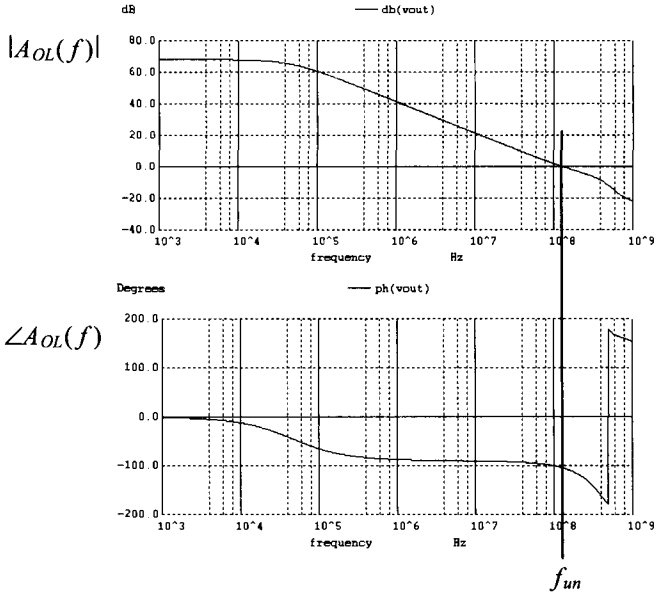


Figure 24.30 Open-loop response of the op-amp in Fig. 24.29.

This is a good point to remember one of the fundamentals we presented in Ch. 9: that is, for high-speed design, we must use minimum length devices (see Eq. (9.59)). If we reduce all of the length of 2 (100 nm) devices in Fig. 24.29 to length of 1 (50 nm) devices, we can increase the speed of the op-amp. Figure 24.32 shows how the step response in Fig. 24.31 changes when we make this reduction in length. The big problem with using minimum channel lengths is the reduction in gain. Close inspection of Fig. 24.32 shows that the output is never quite pulled up to the correct voltage (it is either slightly below 900 mV or 100 mV). Simulating the AC open-loop response results in a gain of 44 dB (= 159). An output voltage of 900 mV would result in a difference on the inputs of the op-amp of 5.66 mV ($0.9/159 = v_p - v_m$).

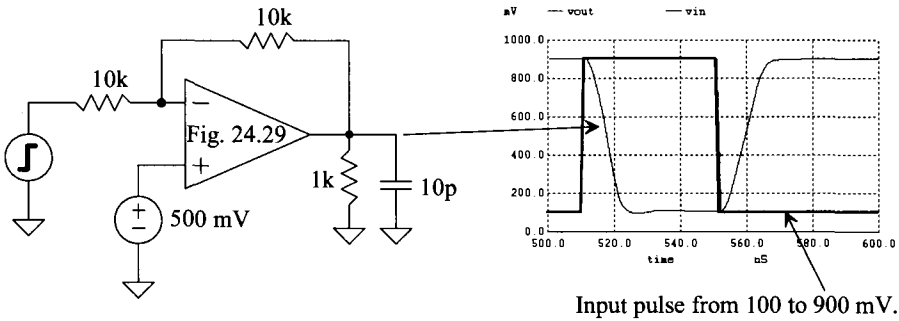


Figure 24.31 Step response of the op-amp in Fig. 24.29 driving 1k and 10 pF.

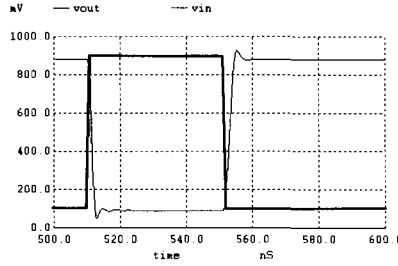


Figure 24.32 Step response of the amplifier in Fig. 24.29 when the lengths of the devices are reduced from 2 to 1 (from 100 nm to 50 nm). The load is still, as seen in Fig. 24.31, 1k and 10 pF.

24.3 The Operational Transconductance Amplifier (OTA)

The operational transconductance amplifier (OTA) can be defined as an amplifier where all nodes are low impedance except the input and output nodes. A simple example of an OTA is the diff-amp with current mirror load seen in Fig. 22.6. *An OTA without buffer can only drive capacitive loads.* A resistive load (unless the resistor is very large) will kill the gain of the OTA.

A sample OTA is shown in Fig. 24.33. Note that the basic op-amp in Fig. 24.2 is not considered an OTA because the drain of M4 is a high-impedance node and not the input or output of the amplifier. As seen in Fig. 24.33, except for the input and output nodes, all nodes have either a gate-drain-connected device or a source connected to them. The terminology “1:K” indicates that M4 and M5 can be sized K times wider ($K > 1$) than the other MOSFETs in the circuit. Assuming that $\beta_1 = \beta_2$, $\beta_{31} = \beta_{41}$, we observe that the current i_{d31} or i_{d41} is given by

$$-i_{d31} = i_{d41} = \frac{g_{mn}}{2}(v_p - v_m) = i_d \quad (24.37)$$

Furthermore, if $\beta_4 = K \cdot \beta_{41} = K \cdot \beta_{31} = K \cdot \beta_3$ and $K \cdot \beta_{51} = \beta_5$, then $i_{d4} = -i_{d5} = K \cdot i_{d41} = -K \cdot i_{d31}$. If the impedance of the capacitor is large compared to $r_{o4} || r_{o5}$, then the output voltage of the OTA is given by

$$v_{out} = 2Ki_d(r_{o4} || r_{o5}) \quad (24.38)$$

and the voltage gain is given by

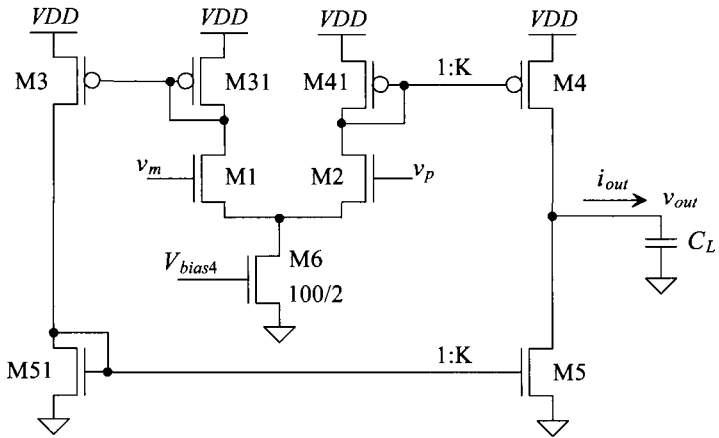
$$A_v = \frac{v_{out}}{v_p - v_m} = K \cdot g_m \cdot (r_{o4} || r_{o5}) \quad (24.39)$$

noting that the noninverting input of the OTA is the gate of M2. While the derivation is interesting, as the name implies, we are interested in the transconductance of OTA. If the impedance of the capacitor is small compared to the OTA output resistance (at higher frequencies), we can write

$$i_{out} = i_{d4} - i_{d5} = 2Ki_d \quad (24.40)$$

The transconductance of the OTA is given by

$$g_{mOTA} = \frac{i_{out}}{v_p - v_m} = K \cdot g_m \quad (24.41)$$



Unless otherwise indicated, parameters from Table 9.2 with biasing circuit in Fig. 20.47.

Figure 24.33 Example of an operational transconductance amplifier (OTA).

Unity-Gain Frequency, f_{un}

To simulate the unity-gain frequency, f_{un} , (where the open loop gain is one) for the OTA, we can use the DC stability scheme seen in Fig. 24.34. We can write (for higher frequencies) the output voltage as

$$v_{out} = i_{out} \cdot \frac{1}{j\omega \cdot C_L} = g_{mn} v_{in} \cdot \frac{1}{j\omega \cdot C_L} \quad (24.42)$$

or

$$\frac{v_{out}}{v_{in}} = \frac{g_{mn}}{j\omega C_L} \rightarrow \left| \frac{v_{out}}{v_{in}} \right| = \frac{g_{mn}}{2\pi f \cdot C_L} \quad (24.43)$$

As before, we define the unity-gain frequency as the point where the open-loop gain is unity

$$\frac{g_{mn}}{2\pi f_{un} \cdot C_L} = 1 \rightarrow f_{un} = \frac{g_{mn}}{2\pi C_L} \quad (24.44)$$

For the OTA in Fig. 24.33 driving a 1 pF load capacitance, we can estimate the unity-gain frequency (with $K = 1$) as

$$f_{un} = \frac{150 \mu A/V}{2\pi \cdot 1 pF} = 24 MHz$$

The simulation results are seen in Fig. 24.34. The location of the pole is estimated using

$$f_{3dB} = \frac{1}{2\pi(r_{o4} || r_{o5})C_L} \quad (24.45)$$

For the OTA in Fig. 24.33, $f_{3dB} = 1.4 MHz$ using the values in Table 9.2. The low-frequency gain is $g_m(r_{o4} || r_{o5}) = 16.65 V/V$ ($= 24.4 dB$).

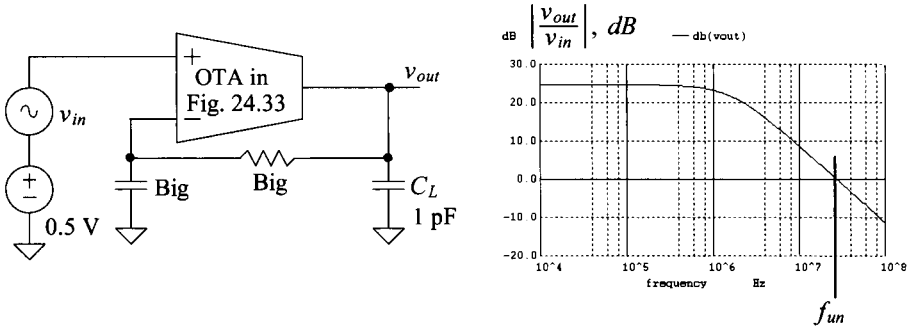


Figure 24.34 Simulating the open-loop response of the OTA in Fig. 24.33.

Note that the maximum value of i_{out} is KI_{SS} (the drain current of M6 multiplied by the increase in widths, K , in the current mirrors M41/M4 and M51/M5). This is important to understand because using an OTA for the first stage of an op-amp results in the same slew-rate limitations of I_{SS} driving C_c that we had when using a diff-amp.

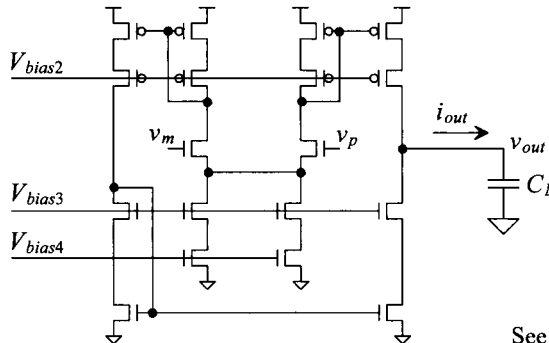
Increasing the OTA Output Resistance

The ideal OTA has infinite output resistance. All of i_{out} flows in the external capacitive load and none flows in the OTA's own output resistance (which is $r_{o4} || r_{o5}$ for the OTA in Fig. 24.33). Towards increasing the OTA output resistance, consider the configuration seen in Fig. 24.35. This topology is based on the one seen in Fig. 24.33 except that now we've cascoded the current mirrors. Note how we've drawn the diff-amp's tail current source as four MOSFETs instead of two with twice the width (this is the way we would lay it out too). The low-frequency gain of the OTA is now increased to

$$A_v = g_{mn} \cdot (R_{ocasn} || R_{ocasp}) \quad (24.46)$$

The 3-dB frequency is now

$$f_{3dB} = \frac{1}{2\pi(R_{ocasn} || R_{ocasp}) \cdot C_L} \quad (24.47)$$



See Table 9.2 and Fig. 20.47.

Figure 24.35 A cascode OTA circuit (higher output resistance).

As discussed before (Sec. 24.2 in the discussion concerning compensating the op-amp), when we get a decrease in the 3-dB frequency and an increase in the low-frequency gain, the effects cancel and the gain-bandwidth product (f_{un}) remains constant. The unity-gain frequency is still given by Eq. (24.44). For the OTA in Fig. 24.35, we can estimate the low-frequency gain, using Tables 9.2 and 20.1, as

$$A_v = (150) \cdot (16.6||4.2) = 500 \text{ (54 dB)}$$

and the 3-dB frequency, driving a 1 pF load, is

$$f_{3dB} = \frac{1}{2\pi(16.6||4.2 \times 10^6)(1 \text{ pF})} = 47 \text{ kHz}$$

Simulation results are seen in Fig. 24.36. The PM is roughly 85°.

An Important Note

Note, in Eq. (24.44), that increasing the load capacitance, decreases the unity-gain frequency, making the OTA more stable. Unlike the two-stage op-amps discussed in the first two sections of this chapter, where the op-amp can become unstable with a large load capacitance, the OTA only becomes more stable with large load capacitance. Also, the PM approaches 90° as the load capacitance increases. The step response of the OTA has a first-order shape (just as in an RC circuit). In many on-chip applications, the load is purely capacitive and so the OTA works great as an “op-amp” in a closed-loop configuration (thus we’ll often call OTAs op-amps). Note that adding a second stage to the basic OTA forms a (two-stage) op-amp like those discussed earlier. We can use the methods already presented to compensate the resulting structure (but, again, driving a load capacitance that’s too large with these two-stage structures can result in instability).

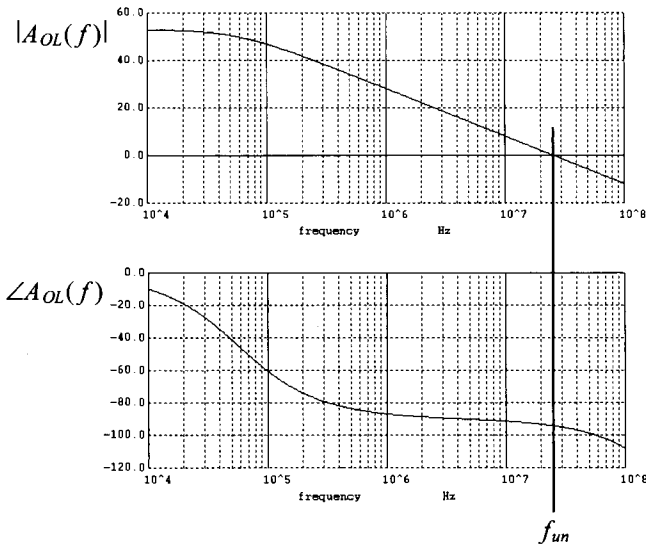


Figure 24.36 The open-loop gain and phase response for the OTA in Fig. 24.35.

OTA with an Output Buffer (An Op-Amp)

If we use the OTA in Fig. 24.35 (or Fig. 24.33) to drive a 1 pF load, we're limited to charging/discharging the capacitor at a rate of

$$\frac{I_{SS}}{C_L} = \frac{dv_{out}}{dt} = \frac{20 \mu A}{1 pF} = \frac{20 mV}{ns} \quad (24.48)$$

For the OTA's output to change from ground to V_{DD} (1 V) requires (just due to slewing) 50 ns. This length of time (the slew-rate limited portion of the settling time) may be fine for many applications. However, for high-speed design, we want faster settling times. Looking at Eq. (24.48), we see that we can decrease the settling time by increasing I_{SS} . However, as discussed in Ch. 9, this causes the gain of the circuit to decrease. To avoid the reduction in gain, we can make a corresponding increase in the size of the devices to keep the overdrive voltage constant. But this is doing nothing more than effectively reducing the size of the load capacitor (the load capacitance becomes comparable in size to the parasitic transistor capacitances).

To speed up the circuit, we may add an output stage, Fig. 24.37. The added stage is a common-source amplifier (with a negative gain indicating **we swap the inverting and noninverting terminals** of the op-amp or else we waste a lot of time). While M7 can turn on and charge a load capacitor quickly, the current source, M8, still limits the rate of load capacitance discharge. For this reason, we've bumped up the current in the output stage to 100 μA . The open-loop simulations driving a 1-pF load are seen in Fig. 24.38. The PM is 45° (too low if the op-amp will be used in a unity follower configuration). Figure 24.39 shows the step response of the op-amp.

We should compare this figure to the response we saw in Fig. 24.31. The op-amp in Fig. 24.31 is driving a heavier load but still has cleaner settling behavior. Further, a comparison of the current pulled from V_{DD} reveals, under the same loading and operating conditions, that the op-amp in Fig. 24.37 pulls more current (see Fig. 24.39). The main difference in these two op-amps is the output buffer used. Let's consider adding a class AB buffer to the OTA in Fig. 24.35 (like the one used in the op-amp of Fig. 24.29). Since we are only driving a 1 pF load, we'll leave the output devices, MON and

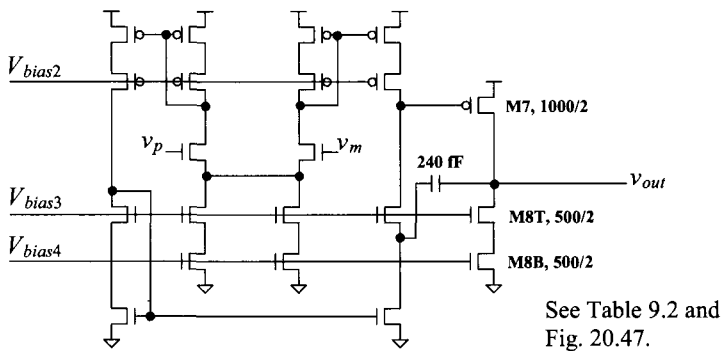


Figure 24.37 A cascode OTA circuit with common-source output buffer (an op-amp). Note the inverting and noninverting terminals are swapped from Fig. 24.35.

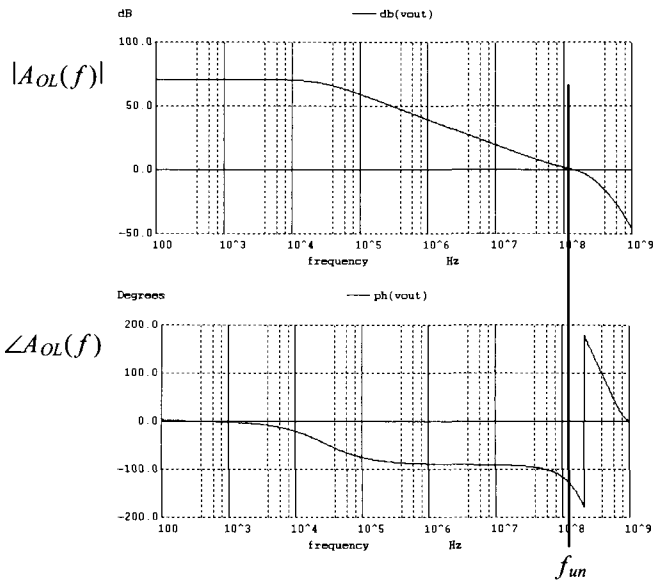
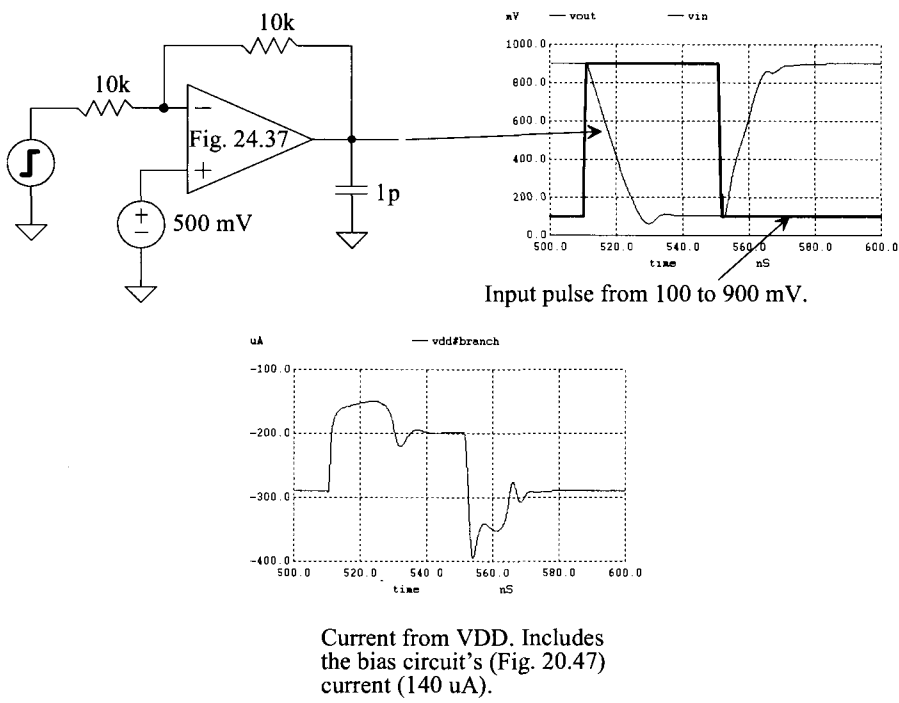


Figure 24.38 Open-loop response of the op-amp in Fig. 24.37 driving a 1-pF load.



Input pulse from 100 to 900 mV.

Current from VDD. Includes the bias circuit's (Fig. 20.47) current (140 uA).

Figure 24.39 Step response of the op-amp in Fig. 24.37 driving a 1 pF.

MOP, at the same size as the other MOSFETs in the circuit (PMOS are 100/2 and NMOS are 50/2). The resulting op-amp is seen in Fig. 24.40. The step response (in the same configuration used to generate Fig. 24.39) is seen in Fig. 24.41. Notice that the current pulled from V_{DD} by the op-amp in Fig. 24.40 is 100 μA less than the current pulled by the op-amp in Fig. 24.37. This lower current is the result of using a class AB output stage.

The performance of the op-amps in Figs. 24.37 and 24.40, with regard to high-speed operation and settling time, won't be quite as good as the topology in Fig. 24.29 (assuming the same biasing conditions, device sizes, and overdrive voltages). To understand why, look at the path, in Fig. 24.40, that the noninverting op-amp input, v_p , takes to get to the op-amp output, v_{out} (that is, M1 - M31 - M3 - M51 - M5 - MON). The delay through this path is longer than the delay, in Fig. 24.29, through M2 - MOP. Further, for high speed, M5 of the op-amp in Fig. 24.40 must turn on (above its quiescent current level) to pull the gate of MON down. For the op-amp in Fig. 24.29, the fact that both the gate of M9T and MOP are driven (in opposite directions) by the diff-amp allows us to use a simple pull-down current (M8) without losing speed. The issues with the cascode diff-amp-based, op-amp topology of Fig. 24.29 are the more limited input common-mode range and the limited voltage swing of node 1. Towards getting better input common-mode voltage range and output swing as well as improved high-speed operation, let's discuss the folded-cascode OTA (see Fig. 22.5).

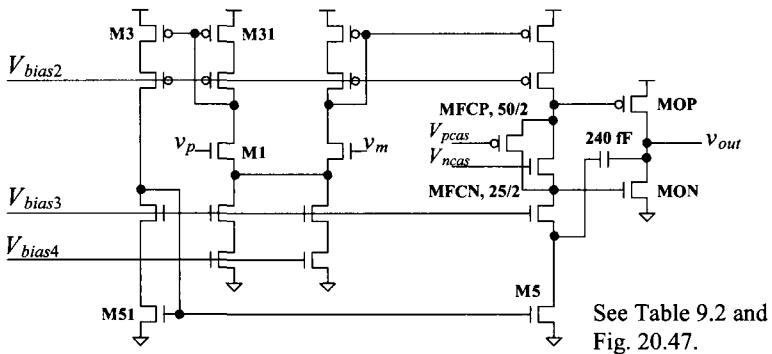
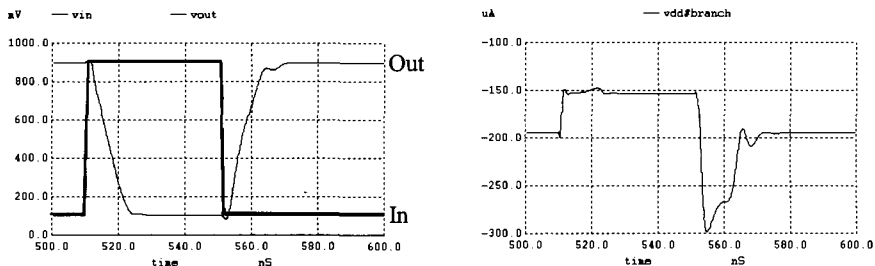


Figure 24.40 Using a class AB output stage with the OTA.



VDD current (includes 140 μA from bias circuit).

Figure 24.41 The step response and current in the topologies seen in Fig. 24.39 using the op-amp in Fig. 24.40.

An example of an NMOS diff-amp-based folded cascode OTA is seen in Fig. 24.42. It is assumed that the reader understands the biasing and operation of the circuits seen in Figs. 20.45, 20.46, and 22.5. To avoid labeling MOSFETs with twice the width, we've drawn the schematic in Fig. 24.42 with MOSFETs in parallel in the branches that supply or sink more current (for example, the diff-amp's tail current). All MOSFETs in Fig. 24.42 conduct 10 μ A of current under equilibrium conditions. Before moving on, it may be a good idea to pause and make sure that the reader understands how the currents sum in this OTA.

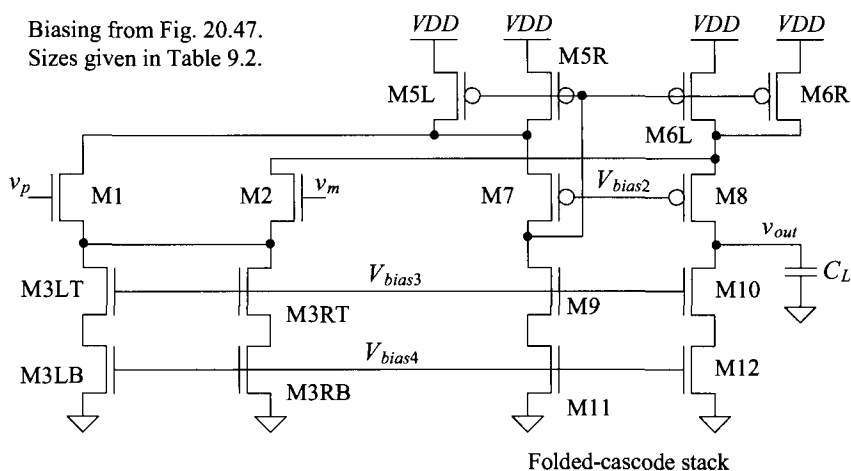


Figure 24.42 A folded-cascode OTA.

The same AC equations presented at the beginning of this section also apply to this OTA. For example, the AC drain currents of M1 and M2 can be written as

$$i_{d1} = -i_{d2} = g_{mn} v_{gs1} = -g_{mn} v_{gs2} \quad (24.49)$$

which, again, indicates $v_{gs1} = -v_{gs2}$. However, we know

$$v_p - v_m = v_{gs1} + (-v_{gs2}) \quad (24.50)$$

so

$$i_{d1} = i_{d5} = i_{d6} = (v_p - v_m) \cdot \frac{g_m}{\gamma} \quad (24.51)$$